Corporate Action Intelligence for BNY: Research Landscape & Actionable Findings for the Capstone Team

TL;DR

- The refined scope — an AI system that reads public documents for voluntary/mandatory-with-options corporate actions, extracts custodian-relevant material information, grounds it in source passages, and maintains an auditable, versioned record — sits in a **genuine white space**: there is abundant research on grounded extraction, confidence calibration, and version-aware RAG, but almost no peer-reviewed academic work applying these techniques *specifically* to corporate-action notice processing, and commercial vendors (Smartstream, Broadridge) are only now moving LLM interpretation "beyond proof of concept."
- The single most directly reusable academic anchor is the 2026 paper **"Grounded Event Extraction from SEC 8-K Filings with a Fine-Grained Taxonomy,"** which independently validates the exact architecture the team should build: schema-constrained output + verbatim-quote grounding + a *separate second-pass* calibrated confidence score (precision rising monotonically from 12% to 96% with the score). Pair it with **VersionRAG** (amendment/change detection), **MAUD** (merger-agreement understanding dataset/benchmark), and **conformal-prediction/selective-prediction** methods for escalation.
- Regulatory tailwinds make the framing timely: **T+1 settlement**, live since May 28, 2024 under amended Exchange Act Rule 15c6-1(a), compresses election windows, and the industry-wide **ISO 20022 migration** plus DTCC's corporate-actions SR2025 cutover (Nov 23, 2025) and modernization program raise the value of clean, structured, auditable event data. Voluntary events are only ~10–15% of volume but carry disproportionate risk, which validates BNY's decision to concentrate there — and The ValueExchange estimates ISO 20022 messaging can lift straight-through-processing efficiency by "over 30% – particularly for complex, voluntary events."

Key Findings

1. **The niche is under-served academically — that is the differentiation.** Despite deep literatures on financial NLP and grounded RAG, there is very little peer-reviewed work on machine-reading corporate-action *notices* (tender offers, rights issues, elections) into structured/auditable records. The space is dominated by patents (e.g., template-driven SWIFT MT56x/ISO 20022 message generation, US 11,720,855) and vendor material, not academic papers. The capstone's neutral, no-recommendation stance plus append-only versioned record-keeping is novel.

2. **A published two-stage grounded-extraction recipe already exists and should anchor the design.** "Grounded Event Extraction from SEC 8-K Filings" (Dolphin et al., 2026) tags 292,984 filings against a 119-type taxonomy; every tag is anchored to a verbatim quote via fuzzy n-gram validation, and a *dedicated second pass* re-grades each quote to produce a calibrated 1–5 quality score. Key finding: inline self-rating saturates and behaves like a binary flag, but the separate re-grade spreads scores and yields precision from 12% (low) to 96% (top). This is a turnkey blueprint for grounding + confidence + escalation.

3. **Amendment/supersession detection has a purpose-built solution.** VersionRAG (Huwiler et al., Oct 2025) models document evolution as a hierarchical version graph, hitting 90% accuracy on version-sensitive QA vs. 58% naive RAG and 64% GraphRAG, and 60% on *implicit* change detection where baselines score 0–10%. This maps directly to "new filing supersedes prior filing" (e.g., SC TO-T/A amendments, extended tender deadlines).

4. **Confidence calibration and abstention are mature enough to build an escalation framework.** Conformal prediction gives distribution-free coverage guarantees; selective-prediction systems (e.g., ExtractConf on the DocILE benchmark) route low-confidence extractions to human review, achieving 99.1% accuracy in the top confidence band. Research repeatedly frames abstention/escalation as the core safety mechanism for high-stakes finance/legal LLMs.

5. **Standards give a ready-made target schema.** ISO 20022 seev.* messages (seev.031 announcement ≈ MT564; seev.033 instruction ≈ MT565; seev.036 ≈ MT566; seev.034 status ≈ MT567) plus DTCC's CA data model and the XBRL US Corporate Actions Taxonomy (covering 90+ event types) provide validated field structures the extraction schema should map to. This directly informs the "minimum useful schema."

6. **Voluntary/complex events are the right risk-concentrated target.** Voluntary events are ~10–15% of corporate actions but carry the highest monetary and reputational risk because a missed election or mis-keyed deadline can mean making good physical assets; mandatory entitlements are allocated by custodians regardless. Per The ValueExchange 2021 survey (with ISSA support), 46% of global corporate-actions event data is processed manually; the same survey (reported via Chainlink) found inefficiencies cost each market participant $3–5 million a year on average and that "Over 45% of brokers have experienced a $1M+ loss due to errors, with some losses costing tens of millions."

7. **Commercial products exist but leave gaps.** Smartstream (Smart Actions), Broadridge, FIS, SimCorp, SmartStream, S&P Global, TCS BaNCS, Vermeg and data vendors (Bloomberg, ICE/EDI, LSEG) automate CA lifecycles. Smartstream states it "has moved beyond proof of concept, applying AI to review narrative

documentation, summarise event descriptions, and identify conditional language within lengthy issuer announcements," with Product Manager Adam Cottingham describing the shift as the "ability to move from deterministic classification to learned, probabilistic classification." None publicly emphasize an academic-style, source-grounded, versioned, neutral summary as the deliverable — the capstone's differentiator.

8. **Regulatory/industry momentum makes the work timely.** T+1 settlement, DTCC's multi-year modernization/cloud/AI program, and the ISO 20022 migration all raise the premium on clean structured CA data. Per DTCC's SR2025 Implementation notice, "DTC will be converting its ISO20022 Corporate Action Messaging to the SWIFT SR2025 version on November 23, 2025," with client testing Sept 8–Oct 31, 2025; SWIFT SR2026 goes live Nov 14, 2026.

Details

Theme 1 — LLM/NLP structured extraction from financial filings (2023–2026)

The strongest, most directly transferable work is **"Grounded Event Extraction from SEC 8-K Filings with a Fine-Grained Taxonomy"** (Dolphin, Dursun, Blankenship, Adams, Pike; arXiv 2607.08346, Jul 2026, from Massive.com). Its two-stage architecture — (1) schema-constrained extraction with self-correcting retries + fuzzy n-gram verbatim-quote grounding, (2) a separate re-grading pass producing a calibrated quality score — is essentially the reference implementation for BNY's grounding + confidence + escalation requirements. The finding that the confidence score is only calibrated when assigned in a *dedicated second pass* (an inline self-rating placed three-quarters of tags at the top score and behaved like a binary flag) is a concrete, testable design decision. The paper also observes that many of the most material disclosures — clinical results, merger completions, regulatory decisions — hide in the "Other Events" catch-all (Item 8.01), reinforcing that classification alone is insufficient and narrative reading is required.

Complementary tooling: LlamaIndex **LlamaExtract** and Tensorlake's **Structured Extraction API** provide field-level citation/source-linking for long filings; both illustrate production patterns for "every extracted value links to its source location." An open-source example, the **corp-actions-scanner** (GitHub, leyremt) parses SC TO-I/TO-T/SC 13E3 tender/exchange filings from EDGAR — useful as a starting harness, though it currently uses heuristic regex for offer-price extraction (a known brittleness the LLM approach would replace). Broader financial-NLP context: FinBERT/transformer surveys, and agentic GraphRAG for unstructured financial data.

Theme 2 — Corporate action standards & schemas

- **ISO 20022 seev.*** structured messages are the canonical target: seev.031 (CA notification), seev.033 (instruction), seev.034 (instruction status), seev.035 (event

processing status), seev.036 (movement confirmation), seev.037 (reversal), seev.039 (cancellation), seev.044 (movement preliminary advice). DTCC publishes MT↔MX mappings (seev.031≈MT564, seev.033≈MT565, seev.036≈MT566, seev.034≈MT567). A seev.031 event is a structured model with explicit event type, separated key dates (record/ex/payment), and linked cash entitlements/currencies/conditions — enabling automated completeness validation.

- **DTCC** uses ISO 20022 for Announcements, Entitlements/Allocations, Instructions and Meetings; its CA 20022 Service anchors events on a **Corporate Action Identifier (CAID)** and the **Official Corporate Action Event Identification (COAF)**, and serves "approx. 1.3 million active securities" (dtcc.com/asset-services/corporate-actions-processing). DTCC's **LENS** (Legal Notice System) provides original, unaltered notice documents — a natural "source of truth" input feed for the capstone.
- **XBRL US Corporate Actions Taxonomy** (from the 2009 DTCC/SWIFT/XBRL US initiative; ADR pilot with BNY Mellon, Citi, JP Morgan, GlobeTax) captures the *issuer's* view of US corporate actions and was built on DTCC and ISO 20022 structures with SMPG/ISITC input. It is highly relevant because it targets tagging *at issuance from documents* — exactly the capstone's input side — and originally covered ~94 event types. XBRL International later chartered a global Corporate Actions Working Group to extend this, tightly integrated with ISO 20022.
- **ISITC** and **SMPG** provide element/configuration recommendations feeding these models.

Takeaway: the team's schema should be a pragmatic subset that maps cleanly onto seev.031 fields (event type, CAID, securities/ISIN/CUSIP, options with option numbers, key dates, rates/prices, conditions, narrative) while adding grounding citations and confidence — fields the ISO messages do not natively carry.

Theme 3 — Source grounding / citation-grounded extraction & confidence

Grounding techniques: verbatim-quote anchoring with fuzzy n-gram validation (8-K paper), post-hoc span/offset citation extraction (claim-to-span mapping, which sidesteps "context rot" where citation accuracy degrades over long generations), and citation-aware chunking with spatial anchors (Tensorlake). Benchmarks for citation quality: **ALCE**, **LongCite/LongBench-Cite**, **RAGAS**, **ARES**; finance-specific: **FinRAGBench-V** (visual citation), **FinDER**, **FinQA/ConvFinQA**. Confidence/calibration: internal-state probes outperform token-logprob/P(True) baselines for detecting hallucinations in financial QA (reported 0.68–0.77 AUROC among "confident" answers vs. 0.55–0.63 for baselines, noting 5–19% of unanimous-confident FinQA answers are wrong); prompt-based vs. fine-tuning calibration surveys; and conformal methods wrapping black-box LLMs for guaranteed coverage without retraining.

Theme 4 — Amendment detection / temporal reasoning / document diffing

VersionRAG is the headline result (90% version-aware accuracy; 60% implicit-change detection; 97% fewer indexing tokens than GraphRAG), formalizing three query types — content retrieval, version retrieval, and change retrieval — each needing distinct strategies. Adjacent: legal statutory temporal-reasoning datasets (pre-amendment questions requiring the *older* version), DACL's native clause versioning with validity windows and a structured audit trail (temporal adjacency, non-overlap, only the latest clause open-ended), and USPTO patent art distinguishing "temporal changes" (against the latest version across a timeline) from "amendment changes" (between adjacent versions). For the capstone this supports an append-only store where each new filing is diffed against the prior authoritative version and material deltas (changed deadline, revised ratio, new condition) are surfaced rather than overwritten.

Theme 5 — Escalation / uncertainty quantification / abstention

Core references: conformal prediction and **conformal risk control** for structured generation; **selective prediction** (coverage–risk trade-off, AURC metric); **ExtractConf** (DocILE, 55 fields, 26% natural failure rate; 0.928 ROC AUC and top-band 99.1% accuracy) as a document-extraction-native selective-prediction system; abstention surveys (AbstentionBench) and the caution that abstention can be a *prompt artifact* rather than genuine uncertainty (important for designing escalation triggers that reflect real uncertainty rather than a triggered "Unknown" slot). Design implication: compute a per-field confidence, set risk-controlled thresholds that bound the error rate among auto-processed events, and escalate whole events when any material field falls below threshold or when sources conflict.

Theme 6 — AI for tender offers, mergers, rights issues vs. simple events

- **MAUD (Merger Agreement Understanding Dataset)**, Wang et al., EMNLP 2023 (arXiv 2301.00876), from The Atticus Project: **152 public merger agreements**, **47,457 annotations** (8,226 deal-point text annotations + 39,231 QA examples), built on the ABA's 2021 Public Target Deal Points Study; annotated by 36 volunteers; **included in LegalBench** (as maud_* tasks). Multiple-choice reading comprehension over deal points (conditions to closing, MAE definitions, remedies) — directly relevant to the "complex narrative" merger/tender documents in scope. Finetuned baselines struggle on the hardest categories, evidence that complex-narrative events are genuinely harder.
- **CUAD** (510 contracts, 13,000+ expert clause annotations, 41 categories, sourced from EDGAR) and **ContractNLI** (~607 NDAs, document-level NLI with multi-span evidence) are the leading clause-extraction/NLI datasets; **ContractEval** benchmarks open vs. proprietary LLMs on CUAD clause-level risk (finding proprietary models lead, and that open models over-emit "no related clause" — a false-abstention failure mode worth watching).
- Why voluntary/complex events are harder is well-documented operationally (Finextra, Oxera, Advent, SIX): they hinge on detailed narrative terms, options, conditions, and deadlines with unforgiving windows, versus mandatory cash dividends which are structurally simple and auto-allocated. Oxera's study explicitly analyzes rights issues and takeovers as archetypal voluntary events with intermediary risk along the chain.

Theme 7 — Open-source tools, benchmarks, datasets for a starting corpus

- **SEC EDGAR full-text search** + APIs; **EDGAR-CORPUS**; the **Stanford EDGAR Filings Dataset (SEFD-v1)** (152B-token snapshot; EDGAR-OCR and EDGAR-Forecast benchmarks) for layout-faithful filings; **FinQA/ConvFinQA/DocFinQA**, **FinanceBench**, **FinDER**, **FinAgentBench** (agentic retrieval over 10-K/10-Q/8-K/DEF-14A/transcripts, 2023–2024), **FiNER-139/FiNER-ORD** and **FinTagging** (XBRL tagging), **TAT-QA** (table+text). For legal-style narrative: MAUD, CUAD, ContractNLI, LegalBench.
- The team can bootstrap a bespoke corpus from EDGAR: SC TO-T/TO-I (tender), SC 14D9 (recommendation), S-4/424B (exchange/rights, prospectuses), DEFM14A (merger proxies), 8-K item 1.01/8.01, and their **/A amendments** — a natural gold set for extraction + amendment detection.

Theme 8 — Industry / vendor landscape and gaps

Celent profiles the CA processing market: Broadridge, FIS, Finastra, SimCorp, SmartStream, S&P Global, TCS BaNCS, Vermeg, plus data providers Bloomberg, EDI, Gresham, LSEG, and specialist Wolters Kluwer; Celent notes vendors are now offering managed data services to combat fragmentation. Broadridge's platform emphasizes "golden copy" data scrubbing from multiple sources, notifications, elections capture with validation, ISO 20022-level integration with DTC, and AI for exception resolution. **Smartstream Smart Actions** covers all mandatory/mandatory-with-choice/voluntary ISO event types within a "fully governed, auditable workflow" and has publicly applied LLMs to interpret narrative issuer documentation. Chainlink and others are exploring blockchain "golden records." **Gap:** vendors focus on lifecycle automation and STP; none publicly center a *source-grounded, versioned, neutral, confidence-scored event summary* as the core research artifact — the capstone's differentiation.

Theme 9 — Agentic architectures with audit trails

LangGraph is the natural orchestration substrate: graph-based workflows with checkpointing, conditional edges, and native human-in-the-loop **interrupt** nodes for escalation (used in production by, e.g., Ally's tech labs). Design patterns from the literature: **verification/self-critique loops** (Perception-Planning-Action-Reflection with HITL escalation, from the Agentic RAG SoK), **multi-agent reflexion** (separate persona-critics + a judge to avoid single-agent echo chambers), and **auditability** best practices (TRiSM for Agentic AI) — comprehensive per-decision logging with timestamps, agent identity, prompt/context, and rationale, plus append-only/tamper-evident ("chain of custody") logs aligned to NIST and ISO/IEC 42001 (event hashes, non-repudiation). This maps cleanly to BNY's audit/versioning requirement: a supervisor agent orchestrating intake→extraction→grounding-check→diff/amendment→confidence-scoring→escalation, with every step logged immutably.

LlamaIndex Agentic Document Workflows offer a meta-agent pattern for coordinating sub-agents over document sets.

Theme 10 — Regulatory / industry trends

- **T+1**: US settlement has been live since May 28, 2024 under amended Exchange Act Rule 15c6-1(a), compressing timelines and elevating the cost of slow/erroneous CA processing (DTCC's T+1 testing explicitly included corporate-actions scenarios).
- **ISO 20022 migration**: the SWIFT MT/MX coexistence period for cross-border payments ended Nov 22, 2025 (with residual translation services persisting); corporate-actions messaging follows the annual Standards Release cadence (DTCC CA cutover to SR2025 on Nov 23, 2025; SWIFT SR2026 Nov 14, 2026). Firms increasingly receive mixed MT564/seev.031 feeds and must normalize early or face parallel logic and manual reconciliation.
- **DTCC modernization**: a multi-year platform/cloud transformation with explicit AI-agent adoption; legacy connectivity/format decommission dates are published for the settlement (Q3 2027) and DRS/DWAC (end-2026) streams, though a corporate-actions-specific legacy sunset date was not found in public DTCC documents. Europe's ECB/AMI-SeCo roadmap targets full ISO 20022 for corporate actions (ISO 15022 and proprietary formats decommissioned) by end-2030.
- **Data-quality economics**: 46% of CA event data processed manually (The ValueExchange 2021/ISSA); $3–5M per-participant annual inefficiency and 45%+ of brokers reporting $1M+ error losses — motivating the business case, alongside The ValueExchange estimate that ISO 20022 can yield "STP efficiencies of over 30% – particularly for complex, voluntary events."

Recommendations

Stage 1 — Literature review & schema (weeks 1–3). Anchor the lit review on: the 8-K Grounded Event Extraction paper (grounding + two-pass confidence), VersionRAG (amendment detection), MAUD/CUAD/ContractNLI (complex-narrative benchmarks), and conformal/selective-prediction papers (escalation). Define the v1 schema as a seev.031-aligned subset (event type, CAID, issuer, ISIN/CUSIP, options+option numbers, record/election/payment dates, financial terms, conditions/restrictions) **plus** grounding citations, per-field confidence, and exception flags. *Benchmark to change plan:* if the seev.031 field set proves too heavy for narrative tender offers, prune to a "custodian communication minimum."

Stage 2 — Corpus & extraction prototype (weeks 3–6). Build a gold corpus from EDGAR (SC TO-T/TO-I + amendments, SC 14D9, S-4/424B, DEFM14A, relevant 8-Ks) across 15–25 events. Implement the two-stage extractor (schema-constrained output + fuzzy n-gram quote grounding, then a separate re-grading pass for calibrated confidence). *Threshold:* target ≥90% precision at the top confidence band on material fields before proceeding; if grounding coverage <95%, iterate chunking/citation before adding features.

Stage 3 — Amendment detection & versioned store (weeks 6–9). Implement an append-only event record; on each new filing, diff against the prior authoritative version (VersionRAG-style change classification: content vs. version vs. change queries) and surface material deltas (changed deadline, revised ratio/price, new condition) without overwriting. *Threshold:* correctly flag the large majority of injected synthetic amendments (extended deadlines, price bumps) in a held-out test.

Stage 4 — Escalation framework (weeks 9–11). Add risk-controlled thresholds (conformal/selective prediction) that escalate an event when any material field is low-confidence or when sources conflict. Log every decision immutably (agent, prompt, rationale, source spans). Enforce the neutral stance: the system surfaces options/conditions/deadlines but never recommends an election. *Threshold:* on the gold set, auto-processed (non-escalated) events should hit a pre-set precision floor (e.g., ≥95%); tune the escalation threshold to hold that floor.

Stage 5 — Orchestration & demo (weeks 11–13). Wrap the pipeline in LangGraph with HITL interrupt nodes and immutable logging; demonstrate downstream consumption against synthetic positions. Explore the stretch MCP/reusable-agent-pattern goal only after the core deliverable is solid.

Evaluation framework (build early, use throughout): score in BNY's stated priority order — (1) extraction accuracy (field-level precision/recall/F1), (2) completeness of material information (recall of a checklist of custodian-required fields per event type), (3) source grounding (% fields with valid, human-verified verbatim span), (4) amendment detection (precision/recall on supersession events), (5) escalation appropriateness (selective-prediction AURC / coverage-risk). Treat processing speed as a reported-but-secondary metric.

Caveats

- **Recency and venue of key sources.** The most on-point extraction paper (arXiv 2607.08346) and VersionRAG (arXiv 2510.08109) are recent preprints; VersionRAG is validated on *technical* documentation, not filings, so its ~90% figures may not transfer directly to corporate actions — the team should re-benchmark on filings.
- **Some statistics are survey-based, secondary, or dated.** The "46% processed manually," "$3–5M per participant," and "45%+ of brokers with $1M+ losses" figures come from The ValueExchange 2021/ISSA survey (relayed via SIX and Chainlink); the "~10–15% of events are voluntary" and "~500,000 elective instructions/year at DTCC" figures are from trade publications, not primary DTCC data — cite as indicative, not authoritative.
- **No corporate-actions-specific DTCC legacy-format sunset date** was found in public documents; the Q3 2027 (settlement) and end-2026 (DRS/DWAC) dates apply to other Transformation streams, and end-2030 is the European (AMI-SeCo) target. Verify with BNY's own DTCC liaison.

- **Vendor claims are marketing.** Smartstream/Broadridge LLM capabilities are described in their own materials and should be treated as directional, not independently verified.
- **The academic-gap claim is a strength but also a risk:** scarce prior work means fewer ready benchmarks for the exact task; the team will need to build its own gold set and evaluation, which is effort-intensive but is itself a contribution.
- **No universal authoritative-source hierarchy exists**, consistent with BNY's scoping; the system must therefore preserve every source version rather than assume one canonical feed, and DTCC's LENS (unaltered notices) plus CAID/COAF identifiers offer a practical backbone for linking versions.